TechBiz Academy

MARYAM REHAN

Cyber Security Internship Report

Week 4 · Task 12 — SIEM Dashboards & Alerts

1. Introduction

SIEM (Security Information and Event Management) helps collect, monitor, and analyze security events. Dashboards and alerts help security analysts identify suspicious activities.

2. Objective

- Monitor failed login attempts.
- Convert search results into visualizations.
- Create a dashboard with two panels.
- Configure an alert for suspicious login activity.
- Understand alert testing.

3. Failed Login Monitoring

Failed login events are monitored to identify repeated authentication failures and possible brute-force activity.

4. Dashboard Creation

Two dashboard panels are used:

- Panel 1: Failed Logins Over Time (Line Chart).
- Panel 2: Top Source IP Addresses (Table).

5. Alert Configuration

An example alert rule uses the following condition:

- Event: Failed Login Attempts
- Threshold: More than 5 attempts
- Time Window: 10 minutes

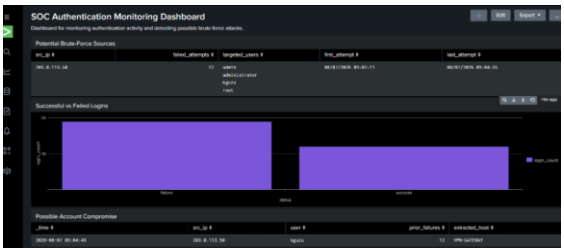
6. Alert Testing

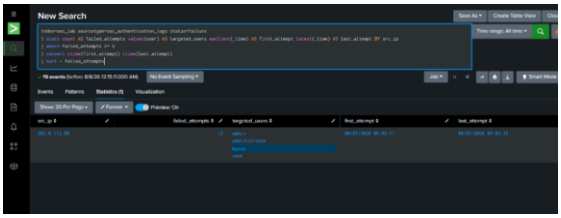
Alert testing is used to check whether the configured rule detects suspicious login activity and generates an alert.

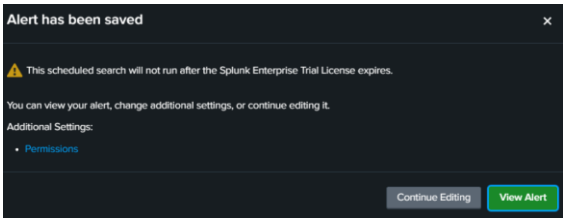
7. Conclusion

This task explains how SIEM dashboards and alerts support security monitoring. Visualizations make login activity easier to understand, while alert rules help identify potentially suspicious authentication behavior.

8. Screen Shoots



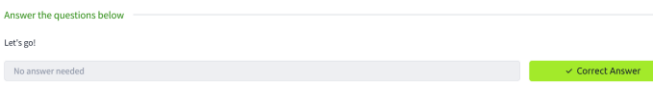




TRY HACK ME



T1



T2

Answer the questions below

What is the process of linking data from multiple sources to identify relationships between individual events called?

Correlation ✓ Correct Answer

What is the process of collecting and storing log data from multiple systems and sources into a single, unified location for easier analysis called?

Centralisation ✓ Correct Answer

T3

Answer the questions below

What is the process of converting logs from different formats into a single format for easier analysis in a SIEM?

Normalisation ✓ Correct Answer

Which log source type can be used to detect the execution of a malicious script?

Host-Based ✓ Correct Answer

T4

Answer the questions below

Which IP address was the connection established with?

10.10.114.80 ✓ Correct Answer

Which process initiated this suspicious connection?

SharePoint.exe ✓ Correct Answer

What is the MD5 hash of the malicious process from the previous question?

770D14FFA142F09730B415506249E7D1 ✓ Correct Answer

What is the name of the scheduled task that was created on the system?

Office365 Install ✓ Correct Answer

T5

Answer the questions below

What was the timestamp of the remote-ssh account creation?
Answer Format Example: 2025-01-15 12:30:45

2025-08-12 09:52:57 ✓ Correct Answer

Which user successfully escalated their privileges to root prior to the action from the first question?

jack-brown ✓ Correct Answer

From which IP address did the user from the previous question successfully log in to the system?

10.14.94.82 ✓ Correct Answer

How many failed login attempts occurred prior to this successful login?

4 ✓ Correct Answer

Which port is the persistence mechanism configured to connect to?

7654 ✓ Correct Answer

T6

Answer the questions below

Which URI path had the highest number of requests?

/wp-login.php ✓ Correct Answer

Which IP address was the source of the activity?

10.10.243.134 ✓ Correct Answer

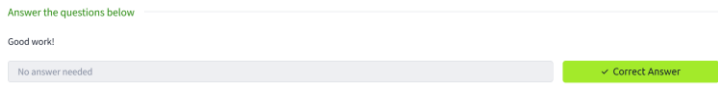
How can this activity be classified?

Brute Force ✓ Correct Answer

Which tool did the threat actor use?

WPScan ✓ Correct Answer

T7



Room completed: 100%

1. Introduction

SIEM (Security Information and Event Management) is a security solution used to collect, monitor, and analyse logs from different sources. It helps security analysts detect suspicious activities, investigate incidents, and respond to potential threats.

2. Objective

The objective of this room was to understand different log sources and learn how to analyse Windows, Linux, and web application logs using Splunk.

3. Task 1: Introduction

This task introduced the purpose of SIEM and its role in security monitoring and incident investigation.

4. Task 2: Benefits of SIEM for Analysts

SIEM provides centralised log collection, correlation, and normalisation. These features help analysts investigate security events from multiple sources in one place.

5. Task 3: Log Sources Overview

Different log sources provide useful information about system and network activities.

- Windows Logs: Help monitor login attempts, processes, and system events.
- Linux Logs: Authentication logs (auth.log) record login and privilege-related activities. System logs (syslog) record general system events.
- Web Logs: Access logs help identify suspicious requests, brute-force attempts, web shells, and possible DDoS activity.

6. Task 4: Windows Logs

Windows event logs were analysed using Splunk. The investigation focused on suspicious network connections, process execution, file hashes, and scheduled tasks.

7. Task 5: Linux Logs

Linux authentication and system logs were studied.

The investigation covered:

- Successful and failed SSH login attempts.
- Suspicious privilege escalation to root.
- Cron jobs and persistence mechanisms.
- Suspicious scripts and reverse shell activity.

8. Task 6: Web Application Logs

Web access logs were studied to identify suspicious web activity.

The main activities included:

- Brute-force attacks against WordPress login pages.
- Possible web shell activity.
- DDoS activity identified through unusual request volumes and HTTP 503 responses.

9. Conclusion

This training improved my understanding of SIEM and log analysis. I learned how Windows, Linux, and web application logs help security analysts investigate suspicious activities. I also understood the importance of monitoring authentication attempts, process execution, scheduled tasks, and unusual web requests.